

X-ray Spectroscopic Diagnostics of Confined and Eruptive Flares

Using SoLEXS/Aditya-L1

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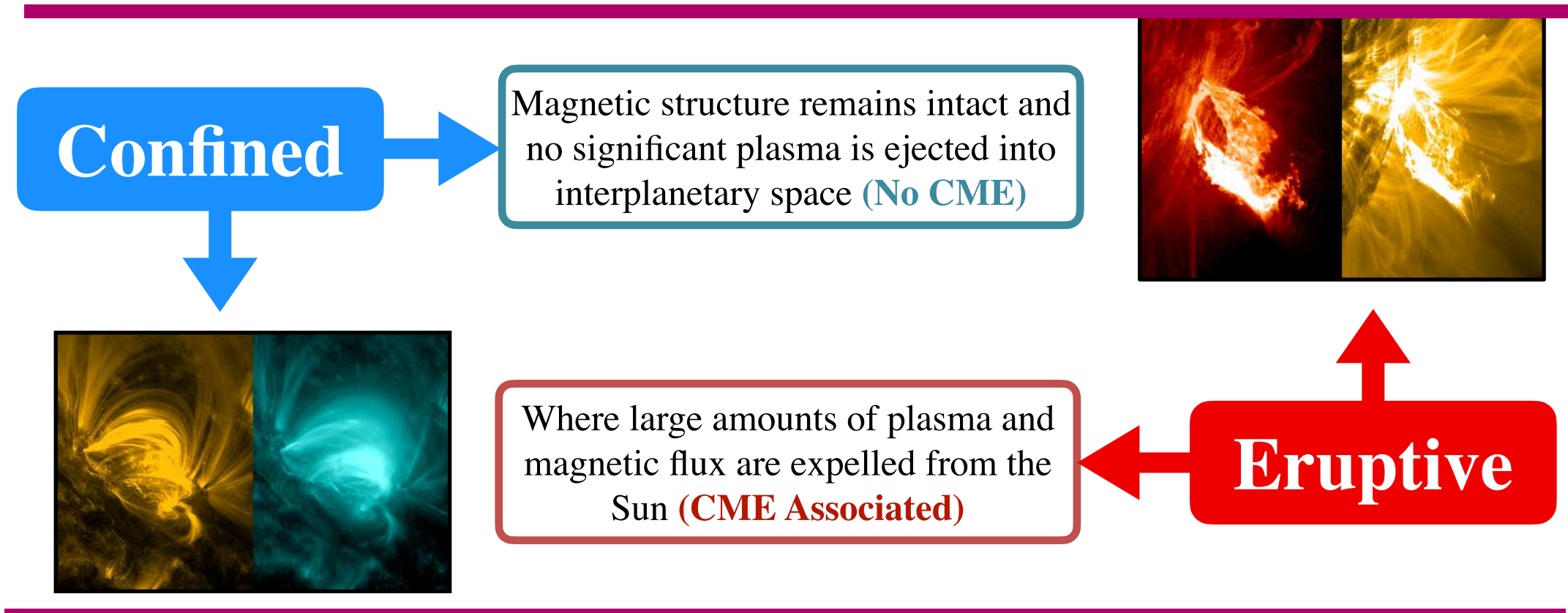


Abstract

Solar flares are broadly classified as confined (no CME) or eruptive (with CME). This study performs a comparative X-ray analysis to quantify the distinct physical characteristics that govern a flare's eruptive potential. We analyze high-cadence soft X-ray spectra of 43 flares observed by the Solar Low Energy X-ray Spectrometer (SoLEXS) onboard Aditya-L1. By fitting the spectra, we derive the temporal evolution of key plasma parameters, including temperature (T), emission measure (EM), and elemental abundances. Our results reveal a clear dichotomy: confined flares are systematically hotter than their eruptive counterparts, while eruptive flares exhibit a significantly higher emission measure, indicating a larger volume of heated plasma. These findings provide direct observational evidence of the different thermodynamic conditions that distinguish confined flares from large-scale eruptive events.

INTRODUCTION

- A solar flare is a **sudden, intense release of radiation** caused by **magnetic energy** release in the Sun's atmosphere.
- Flares generally occur near **sunspots in active regions**, where twisted magnetic field lines reconnect and release energy.
- Flares can be classified as
 1. Confined
 2. Eruptive



Flare Class	Associated X-ray Flux (W/m ²)
A	< 10 ⁻⁷
B	10 ⁻⁷ - 10 ⁻⁶
C	10 ⁻⁶ - 10 ⁻⁵
M	10 ⁻⁵ - 10 ⁻⁴
X	> 10 ⁻⁴

SOLEXS & DATA PRODUCTS

- The **Solar Low Energy X-ray Spectrometer (SoLEXS)** is a soft X-ray spectrometer aboard Aditya-L1, to measure the solar soft X-ray flux to study flares.
- Sun as a Star Spectrometer** (intergrated flux)

Data Products -

- Good Time Intervals Files:** Time periods of valid data collection by the detector
- Light Curve Files:** 1-second-cadence count rates (2–22 keV)
- Pulse Invariant Files:** Calibrated energy spectra recorded every second.

Parameter	Specification
Energy Range	2–22 keV
Energy Resolution	(2.4–22 keV for spectral fitting)
Time Cadence	1 second
Detector	Silicon Drift Detector (SDD)
Type	2 (named SDD1 & SDD2)
Active Area	400 ± 20 mm ²
Thickness	8 µm thick Be
Aperture Area	SDD1: 7.106 mm ² SDD2: 6.106 mm ²
Field of View	SDD1: ±1.8° SDD2: ±1.8°
Calibration Source	Fe-55 with Ti foil
Digital Pulse Processing parameters	Spectral Channel: 2 µs Pulse peaking time (triangular pulse): 0.35 µs
Number of channels in the spectrum	340
Channel Width	1–168 channel: ~47.6 eV 169–340 channel: ~95.2 eV

EVENT SELECTION

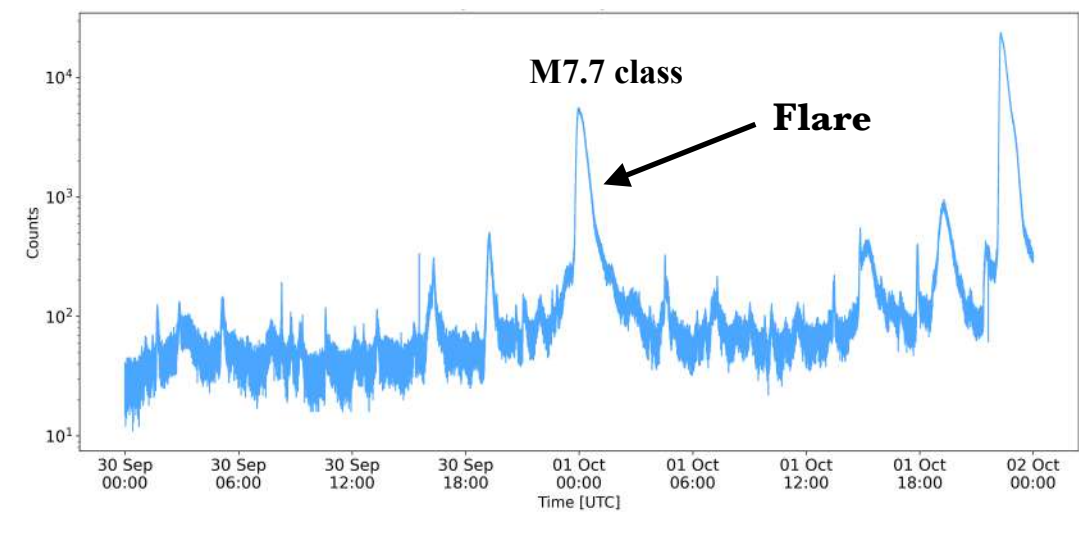
- Flare identification**
 - Selected events from the **GOES Flare Catalog**
 - Time period: **1 July 2024 – 31 July 2025**
- Verification**
 - Cross-checked with **SoLEXS X-ray light curves**
- CME association**
 - Identified using the **SOHO/LASCO CME catalog**

Flare Class	Eruptive	Confined	Total
C-class	4	4	8
M-class	17	7	24
X-class	10	1	11
Total	30	13	43

METHODOLOGY

Data Processing

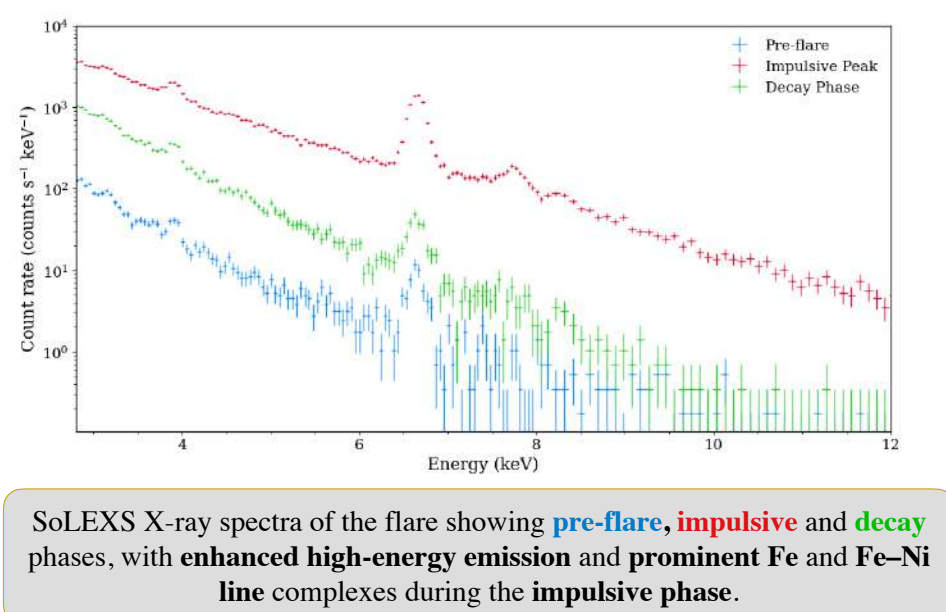
Identify the **Flare** in the **Solexs lightcurve**



Spectra Generation

Spectral Files were generated for each Flare.

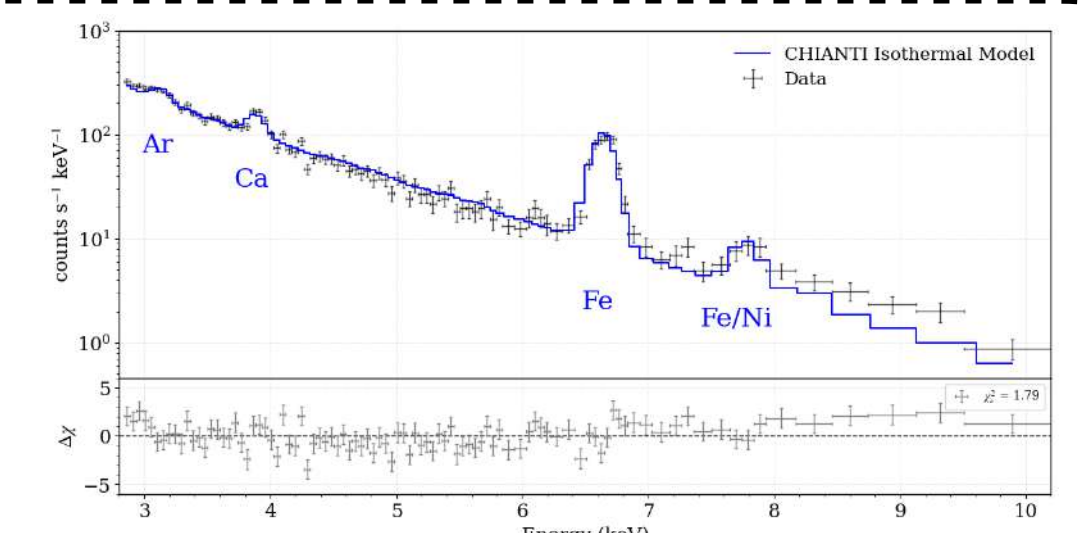
Time bin intervals were optimized to ensure a high **signal-to-noise ratio** and **goodness of fit**.



Model & Free Paramters

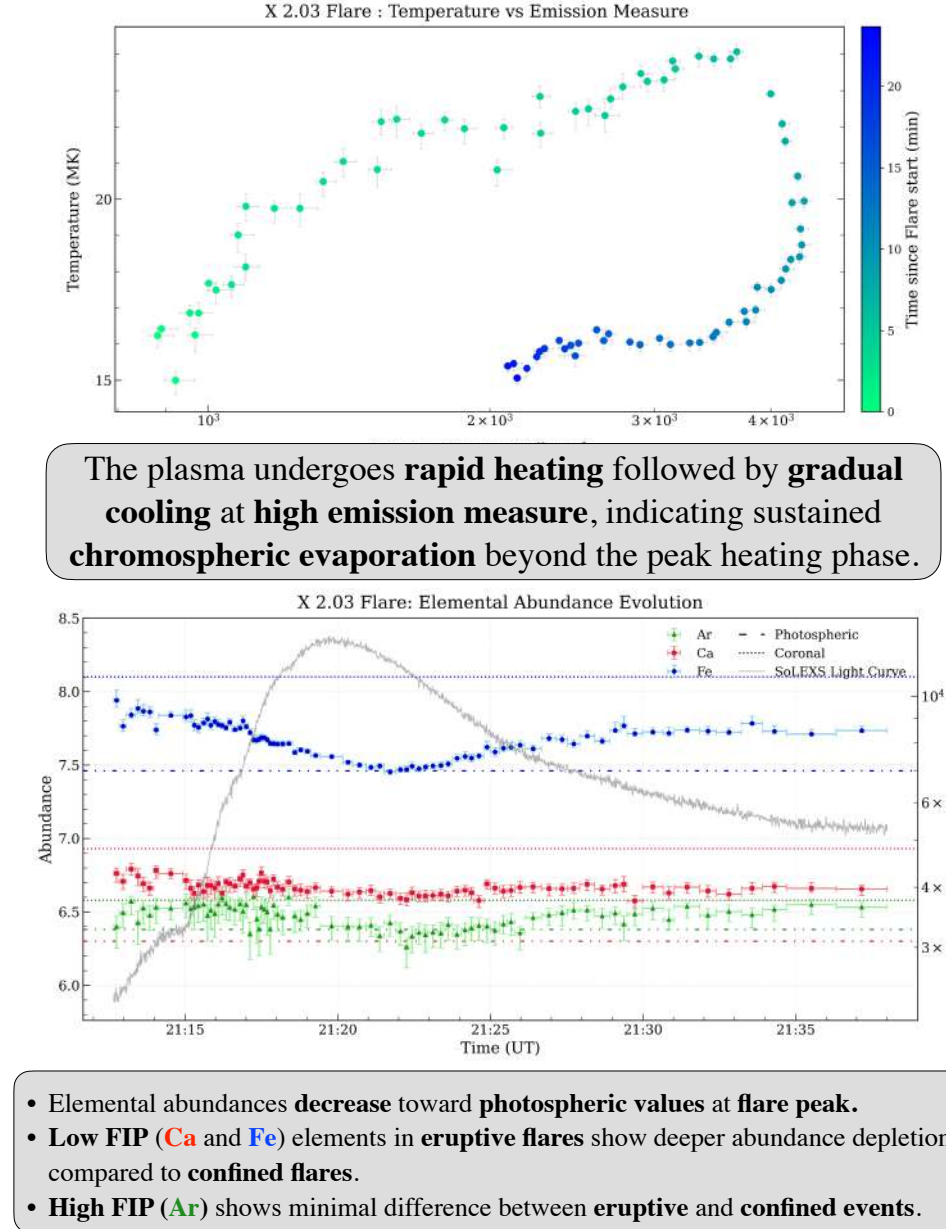
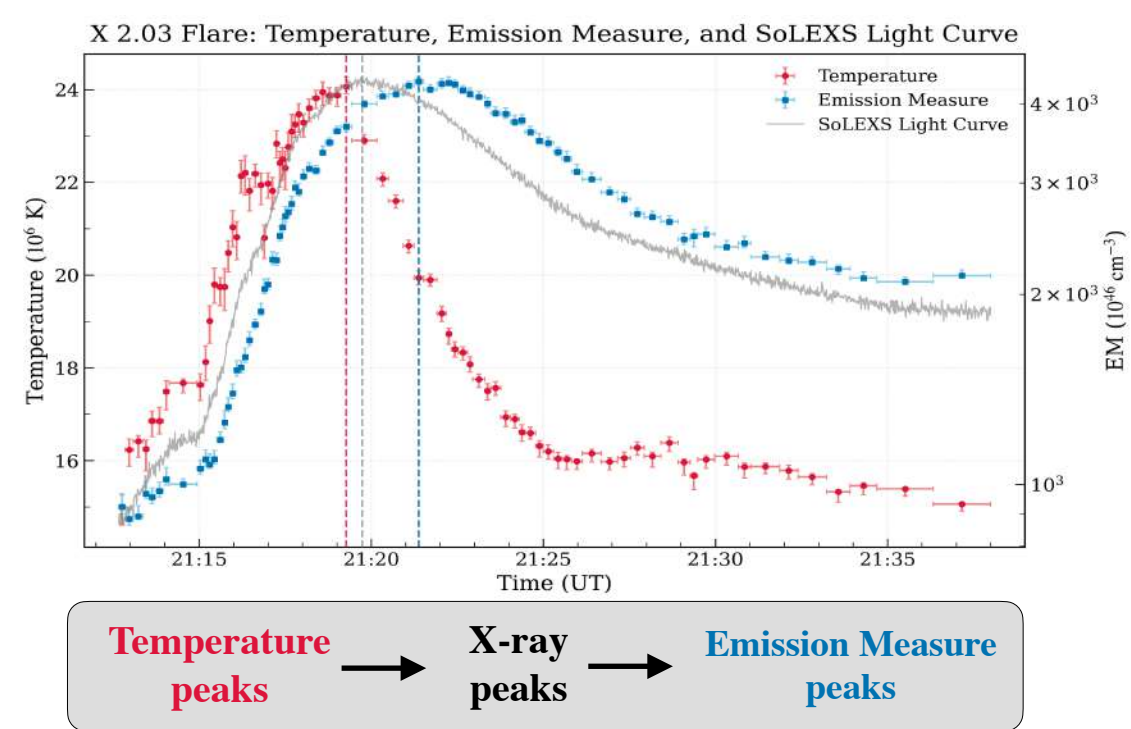
Spectra were fitted with **PyXSPEC** using the **Chianti Isothermal model** (chisoth).

Energy range used: **2.8–12 keV**

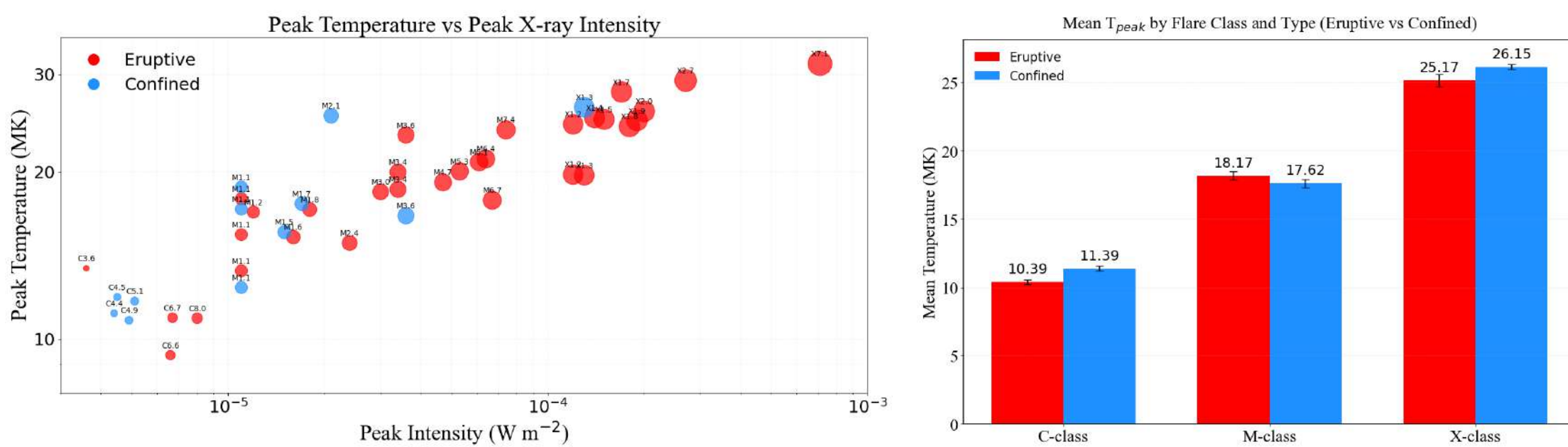


Spectral Fitting

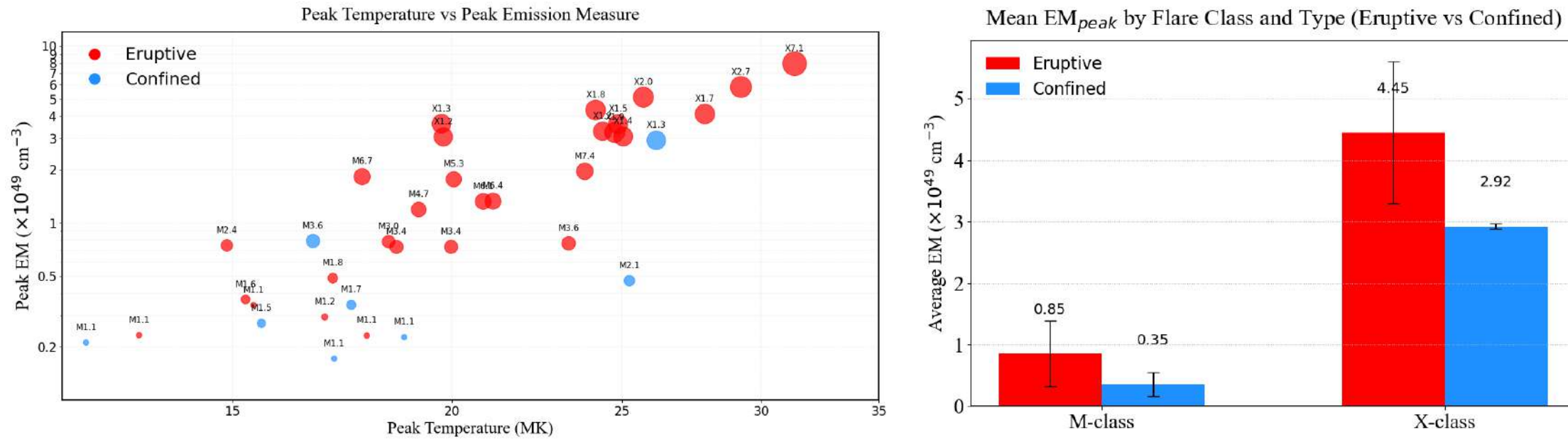
Temperature, Emission Measure, and Abundances of Ar, Ca, Fe, Ni are left free during fitting



RESULTS



- C-Class & X-Class:** **Confined flares** show **higher peak temperatures** than **Eruptive**.
- M-class flares:** **Mixed behavior** — some **confined flares** are hotter, others not.



- Peak Emission Measure **increases** with Peak Intensity (Flare Class).
- Eruptive Flares** have significantly **higher Emission Measure** than **Confined Flares**.

CONCLUSION

- Significant **Time lag** between Peak Temperature, Peak X-ray Flux, and Peak Emission Measure.
- Peak Temperature and Peak Emission Measure **increase** with Peak Intensity.
- Confined Flares may exhibit **higher peak temperatures** than Eruptive Flares
- Eruptive Flares have a **higher EM** than Confined Flares
- Confined Flares show a **more photospheric abundances** of low FIP elements during flare peak.

REFERENCES

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- XSPEC: The First Ten Years**. Astronomical Data Analysis Software and Systems (Arnaud, K.A., 1996.)
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